Pricing, Hedging, and Model Risk

Lecture 13 Notes — Vanilla, Path Dependence, and Stochastic Rates

Goals and logical structure

Pricing is only the beginning of a derivatives workflow:

$$\text{contract} \longrightarrow \text{state variables} \longrightarrow \text{model and price} \longrightarrow \text{hedge} \longrightarrow \text{model-risk audit.}$$

This lecture compares three situations: a vanilla option, a lookback option, and a claim exposed to stochastic interest rates.

1 A vanilla option: the hedge depends on the model

For a European payoff $g(S_T)$, Black–Scholes assumes

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}.$$

The price is

$$v(t,s) = \mathbb{E}_{t,s}^{\mathbb{Q}}[e^{-r(T-t)}g(S_T)],$$

and the delta hedge holds $\Delta_t = v_s(t,S_t)$ shares. In this one-factor complete model, continuous rebalancing replicates the claim.

Example — local volatility changes delta

Under local volatility,

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma_{\mathrm{loc}}(t,S_t)\,\mathrm{d}W_t^{\mathbb{Q}},$$

and

$$v_t + rsv_s + \tfrac{1}{2}\sigma_{\mathrm{loc}}(t,s)^2 s^2 v_{ss} - rv = 0.$$

The formal hedge remains $\Delta = v_s$, but its value and delta depend on the calibrated surface. A hedge is therefore a model-dependent object.

A good vanilla fit is not a guarantee of a good hedge

If the market follows stochastic volatility while the desk uses a local-volatility delta, the option price may initially match market vanillas but the subsequent smile movement can generate systematic hedging P&L. Calibration error, discrete rebalancing, transaction costs, and parameter instability add further sources of error.

Exercise — price versus hedge

Two models match today's vanilla call price but imply different deltas. Can their future hedging P&L differ?

Short solution

Yes. Equality of one initial price does not imply equality of future sensitivities or of model-implied smile dynamics. Hedging P&L depends on the path followed by market risk factors and on the hedge rule.

2 A lookback option: the path becomes state

2.1 The additional state variable

A fixed-strike lookback put may pay

$$M_T - S_T, \qquad M_t = \max_{0\le u\le t} S_u.$$

Knowing S_t alone is insufficient: two paths with equal current spot but different running maxima have different future payoffs. The Markov state is (S_t, M_t).

Definition 2.1 (State augmentation)**.** When a payoff depends on a path statistic, include the statistic as a state variable. Pricing then becomes Markovian in the enlarged state, not necessarily in the stock price alone.

Away from a new maximum, a local-volatility lookback value $u(t,s,m)$ satisfies

$$u_t + rsu_s + \tfrac{1}{2}\sigma_{\text{loc}}(t,s)^2 s^2 u_{ss} - ru = 0, \qquad 0 < s < m,$$

with terminal condition $u(T,s,m) = m - s$. The diagonal $s = m$ requires a boundary condition describing the update of the running maximum.

Path dependence has a computational cost

The new state adds a grid dimension and makes simulation bookkeeping essential. In Monte Carlo, each path must carry its own running maximum; in a PDE, the diagonal boundary must be treated consistently.

Exercise — minimal state

For an arithmetic Asian call with payoff $(T^{-1}\int_0^T S_u\,\mathrm{d}u - K)^+$, what extra state restores Markovianity?

Short solution

Use the running integral $I_t = \int_0^t S_u\,\mathrm{d}u$, so the state is (t, S_t, I_t). The accumulated average alone does not replace the current stock because future accumulation still depends on S_t.

3 Stochastic rates add a hedging factor

If the short rate is random, discounting is random:

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_u\,\mathrm{d}u\right)\middle|\mathcal{F}_t\right].$$

For a stock and a short-rate factor,

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - q)\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t^S, \qquad \mathrm{d}r_t = a(t,r_t)\,\mathrm{d}t + \sigma_r(t,r_t)\,\mathrm{d}W_t^r.$$

The value of a European claim is generally $v(t,S_t,r_t)$. Its hedge exposure includes both v_s and v_r.

Example — what becomes tradable?

Stock delta manages the stock shock. A zero-coupon bond or an interest-rate instrument is needed to manage the rate shock. If sufficiently many independent risk factors cannot be spanned by traded instruments, the market is incomplete and perfect replication is unavailable.

Model risk and market incompleteness are different

Model risk is error from a misspecified or unstable model. Incompleteness is a structural fact: available traded assets cannot span all shocks. Both can produce residual P&L, but their remedies differ.

4 Hedging P&L as a diagnostic

For a short option position, a simple discrete hedging P&L is

$$\Pi_T = V_0 + \sum_i \Delta_{t_i}(S_{t_{i+1}} - S_{t_i}) - \text{option payoff} - \text{funding and costs.}$$

Analyse not only its mean, but also dispersion, tail losses, dependence on spot moves, volatility moves, and rate moves. A model can have small average error while producing unacceptable tail risk.

Practical workflow

1. Identify every payoff-relevant state variable.
2. Calibrate the model to liquid instruments that support hedging.
3. State the hedge instruments and rebalance convention.
4. Backtest and stress-test the distribution of hedging P&L.
5. Report residual risk rather than claiming a perfect hedge by default.

Exercise — final diagnosis

A long-dated lookback option is priced with constant volatility and deterministic rates. Name two material modeling omissions to investigate.

Short solution

The path-dependent payoff can be very sensitive to volatility-surface and smile dynamics, so constant volatility is a key omission. Long maturity also makes stochastic-rate and stock-rate correlation risk potentially material. The correct choice depends on available market instruments and the P&L impact of these risks.